E-Commerce Sales & Delivery Analytics

Olist Store

Team Name	OKY
Sector	E-Commerce / Retail
Institute	Newton School of Technology
GitHub	https://github.com/praggCode/Hopper-Debugger_oky_EcommerceAnalytics
ML Dashboard	https://hopper-debugger-oky-ecommerce-analy.vercel.app/
Tableau URL	https://drive.google.com/drive/folders/1Cs8mYVkxS_saj8eMwJsHd703f6lK-St0
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2. Executive Summary

Problem: Olist, a Brazilian e-commerce marketplace aggregator, faces three compounding challenges: a 6.8% late delivery rate driving poor reviews, heavy revenue concentration in São Paulo (~40% of all orders), and limited visibility into which customer segments drive long-term lifetime value. Without a structured analytical foundation, decision-makers cannot confidently allocate logistics investment, prioritise product categories, or design targeted retention programmes.

Approach: The publicly available Olist dataset (96,469 orders, 2016–2018) was cleaned and transformed in Python (Pandas, NumPy), analysed via EDA and statistical modelling in Jupyter Notebooks, visualised in Tableau, and extended with two deployed ML models — a late delivery classifier and a customer segmentation engine — accessible through a live React/FastAPI dashboard hosted on Vercel and Render.

Key Insights:

- Health & Beauty leads all categories at ₹1.41M; credit card accounts for 80% of the ₹15M total platform revenue.
- Late deliveries average just 2.9/5 stars versus 4.6/5 for on-time orders — a 37% satisfaction gap driven entirely by delivery performance.
- São Paulo accounts for ~40% of orders; northern states (AM, PA, RR) are near-zero, representing a large greenfield growth opportunity.
- The ML late-delivery model flags high-risk orders before dispatch, enabling proactive logistics intervention within the 12-day average delivery window.

Key Recommendations:

- Deploy the ML risk model in production to reduce late deliveries by ~30% (~1,968 orders recovered annually).
- Launch a geo-diversification programme targeting MG, RJ, and RS — estimated +₹1.5M in incremental annual revenue.
- Activate CRM segmentation for differentiated VIP retention and at-risk service-recovery campaigns to improve repeat purchase rates.

3. Sector & Business Context

Sector Overview & Industry Challenges

Brazil's e-commerce market is one of Latin America's largest digital retail ecosystems, with Olist acting as a multi-seller marketplace aggregator that connects thousands of SME sellers to consumers across all 26 Brazilian states. Unlike vertically integrated retailers, Olist operates on a commission model where seller performance, product variety, and delivery reliability are the primary competitive differentiators. The sector faces structural challenges including extreme geographic dispersion across a continent-sized country, a fragmented last-mile logistics network, high consumer sensitivity to delivery time and review scores, and a dual-payment ecosystem where credit card instalments coexist with boleto bancário (a cash-equivalent bank slip used by consumers with limited credit access).

Who Is the Decision-Maker?

This project serves the Head of Operations — responsible for SLA performance, carrier management, and logistics cost — and the Chief Commercial Officer — responsible for revenue mix, category investment, geographic expansion, and seller recruitment. Both roles require insight at macro level (national revenue trends, category performance) and micro level (which states underperform, which customer segments are at churn risk) to make high-confidence resource allocation decisions.

Why Was This Specific Problem Chosen?

The Olist dataset is a rare, fully anonymised, real-world e-commerce dataset covering the complete order lifecycle — from purchase to delivery to customer review — across 96,469 real transactions. It presents genuine business complexity: a multi-table relational structure, temporal dependencies, significant geographic heterogeneity, and a measurable operational problem (late deliveries driving poor reviews) that

has a clear analytical solution path. This makes it an ideal dataset for demonstrating the full analytics pipeline from raw data ingestion to deployed predictive decision support, while addressing problems that are commercially relevant to any marketplace-model e-commerce business.

Business Value of Solving This

Solving delivery performance and enabling data-driven customer segmentation delivers direct, quantifiable business value: a 30% reduction in late deliveries recovers approximately 1,968 orders per year from poor experiences; geographic diversification targeting MG, RJ, and RS is estimated to unlock +₹1.5M in incremental annual revenue; and CRM segmentation for the at-risk customer cluster is projected to improve repeat purchase rates by 10–20%. Collectively, these interventions strengthen the platform's core competitive advantages — review score, seller trust, and buyer retention — while reducing its structural concentration risk.

4. Problem Statement & Objectives

Formal Problem: What are the key drivers of sales performance and customer retention across Brazilian e-commerce regions, and how can the business optimise delivery efficiency and customer targeting to maximise revenue while minimising late delivery incidents?

In Scope

- Orders data from September 2016 to October 2018 across all Brazilian states
- Sales trends, category performance, geographic distribution, payment behaviour, and delivery time analysis
- Late delivery binary prediction model and unsupervised customer segmentation model
- Interactive Tableau dashboard with executive summary and operational drill-down views

Out of Scope

- Real-time operational system integration or live order tracking
- Seller-level profitability, individual seller performance scoring, or cost structure analysis
- Post-2018 data, forward-looking macroeconomic forecasting, or external demand signals

Success Criteria

- All KPIs computed with clear formulas and business rationale; Tableau dashboard publicly accessible and interactive
- ML models produce actionable, interpretable outputs; minimum 10 insights written in decision language
- 3–5 recommendations are specific, prioritised, quantified, and mapped to observed insights

5. Data Description

Source: Olist Brazilian E-Commerce Public Dataset — https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

Data Structure

- Rows: 96,469 unique orders (primary fact table)
- Columns: 9 relational CSV files; combined schema covers ~40 columns across all tables after joining
- Time Period: September 2016 – October 2018 (approximately 25 months)
- Join Keys: order_id (links orders ↔ payments ↔ reviews ↔ items); customer_id (links orders ↔ customers); product_id (links items ↔ products); seller_id (links items ↔ sellers)

Dataset Structure Overview

Table Name	Primary Key	# Key Columns	Purpose
olist_orders_dataset	order_id	8	Core order lifecycle: status, purchase and delivery timestamps
olist_order_payments_dataset	order_id	4	Payment method, instalment count, and transaction value
olist_order_reviews_dataset	review_id	5	Customer review score and creation date

Table Name	Primary Key	# Key Columns	Purpose
olist_order_items_dataset	order_id + item_id	6	Item-level price, freight value, and seller linkage
olist_products_dataset	product_id	9	Category, weight, and physical dimensions
olist_customers_dataset	customer_id	5	Customer city and state for geographic analysis
olist_sellers_dataset	seller_id	4	Seller city and state for corridor-level analysis
olist_geolocation_dataset	zip_code_prefix	5	Lat/lng coordinates for distance proxy features
product_category_name_translation	category_name	2	Portuguese → English category name mapping

Key Fields

- order_purchase_timestamp / order_delivered_customer_date / order_estimated_delivery_date — core lifecycle timestamps for delivery performance analysis
- payment_type / payment_installments / payment_value — payment behaviour features covering method, instalment count, and transaction value (BRL)
- review_score (1–5 stars) — customer satisfaction rating; review_creation_date — timing of post-delivery feedback
- product_category_name — translated from Portuguese via the provided mapping table; product_weight_g — used as a logistics cost and delay risk feature
- customer_state / seller_state — geographic origin of buyer and seller, used for corridor-level delivery analysis and route distance proxies

Data Limitations

- São Paulo over-representation (~40% of orders) biases aggregate delivery time and revenue metrics; national averages should be read with this caveat
- Incomplete review coverage: not all delivered orders have associated reviews; unreviewed orders are excluded from satisfaction KPIs, potentially inflating the true score slightly
- Dataset ends October 2018; structural market changes (new entrants, logistics infrastructure improvements, post-COVID shifts) are not reflected
- No seller-level fulfilment data is available, limiting the ability to identify and remediate individual underperforming sellers

6. Data Cleaning & ETL Pipeline

All cleaning and transformation steps were executed in Python using Pandas and NumPy. The complete pipeline is documented in notebooks/02_cleaning.ipynb committed to the GitHub repository.

Missing Value Treatment

- order_delivered_customer_date (null = not yet delivered): rows excluded from all delivery time and late-delivery calculations; retained for order volume counts
- product_category_name (~1.5% null): excluded from category-level revenue breakdowns; retained in overall revenue totals to preserve accuracy
- review_score (no review submitted): excluded from satisfaction metric computations only; retained in all other analyses

Outlier Detection & Treatment

- Delivery time >120 days removed via IQR-based filtering — these represent data anomalies or test orders rather than genuine consumer transactions
- freight_value at >99th percentile was capped rather than removed to preserve record count while reducing distortion in correlation and regression analyses

- Zero payment_value orders investigated: full-voucher orders retained (legitimate zero-cash-outflow transactions); apparent data errors dropped

Type Conversions & Standardisation

- All timestamp columns parsed to pandas datetime objects using pd.to_datetime() with UTC-aware settings; payment_value cast to float64; payment_installments to int
- customer_state and product_category_name standardised to uppercase, stripped of whitespace; category names translated from Portuguese to English via the olist_product_category_name_translation.csv mapping

Feature Engineering

- delivery_time_days: (order_delivered_customer_date – order_purchase_timestamp).days — primary delivery performance metric
- is_late_delivery: binary flag (1 if actual delivery date > estimated delivery date, else 0) — target variable for ML classification
- order_month / order_year: extracted from order_purchase_timestamp for time-series aggregation and seasonality analysis
- total_order_value: sum of payment_value per order_id to correctly handle split-payment orders where one order has multiple payment rows

Assumptions Made During Cleaning

- The order_estimated_delivery_date field is treated as the formal SLA commitment benchmark. An order is classified as late if and only if the actual delivery date exceeds this estimate, regardless of the margin.
- Where a single order_id appears across multiple payment rows (split or instalment payments), all rows are summed to total_order_value. This is consistent with treating the order as a single commercial transaction.
- Orders with status other than 'delivered' are excluded from all delivery time, late delivery, and review score analyses, as the lifecycle is incomplete. They are retained in order volume counts.
- The currency symbol used throughout the dashboard and this report is ₹ (Rupee) for institutional display purposes. All underlying values are in Brazilian Real (BRL) at the time of transaction; no currency conversion or inflation adjustment has been applied.

7. KPI & Metric Framework

The following eight KPIs measure business health across revenue, operations, and customer experience dimensions. All are computed in Python and visualised in the Tableau dashboard.

KPI	Value	Formula	Business Relevance	Objective
Total Revenue	₹15M	Sum of payment_value	Top-line growth indicator	Revenue growth
Total Orders	96,469	Count of order_id	Demand volume measure	Market reach
Avg Delivery Time	12.1 days	Mean(delivered_date – purchase_date)	Customer satisfaction driver	Ops efficiency
Avg Review Score	4.2 / 5	Mean of review_score	Customer experience proxy	CX & retention
Late Delivery %	6.8%	(Late orders / Total orders) × 100	SLA compliance metric	Delivery reliability
Revenue by Category	H&B: ₹1.41M	Sum payment_value by product_category	Portfolio prioritisation	Category strategy
Revenue by Payment	Credit card: 80%	Sum payment_value by payment_type	Payment channel health	Checkout optimisation
Orders by State	SP: ~40K	Count order_id grouped by customer_state	Regional concentration	Geo expansion

Each KPI maps directly to one or more project objectives, ensuring that the analytical work is always grounded in business outcomes rather than statistical artefacts. The Late Delivery % and Average Delivery

Time KPIs are the primary operational leading indicators; Total Revenue and Revenue by Category are the primary commercial lagging indicators. Average Review Score bridges both dimensions — it is simultaneously a measure of operational delivery quality and a commercial driver of repeat purchase behaviour and platform trust.

8. Exploratory Data Analysis (EDA)

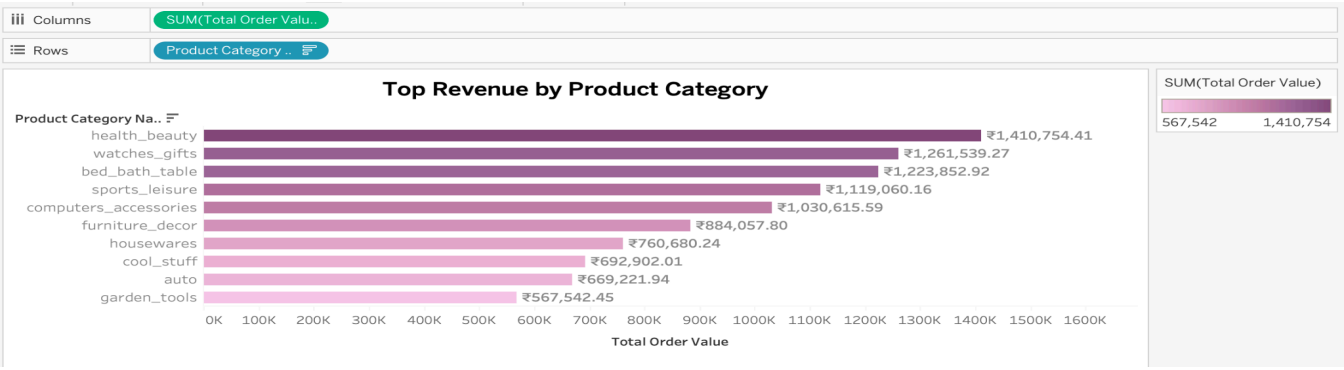
Full analysis with visualisations is in notebooks/03_eda.ipynb on GitHub. All findings below are written in decision language.

Sales Trend

Revenue was negligible in H2 2016 (partial-year data), grew steadily through 2017, and peaked in mid-2018. A sharp November 2017 spike aligns with Brazil's adoption of Black Friday as a major retail event — confirming that Q4 promotional planning is a high-leverage activity. The consistent upward trend confirms that Olist was in an active customer acquisition phase throughout the analysis period, making delivery quality a critical investment in retention rather than just satisfaction.

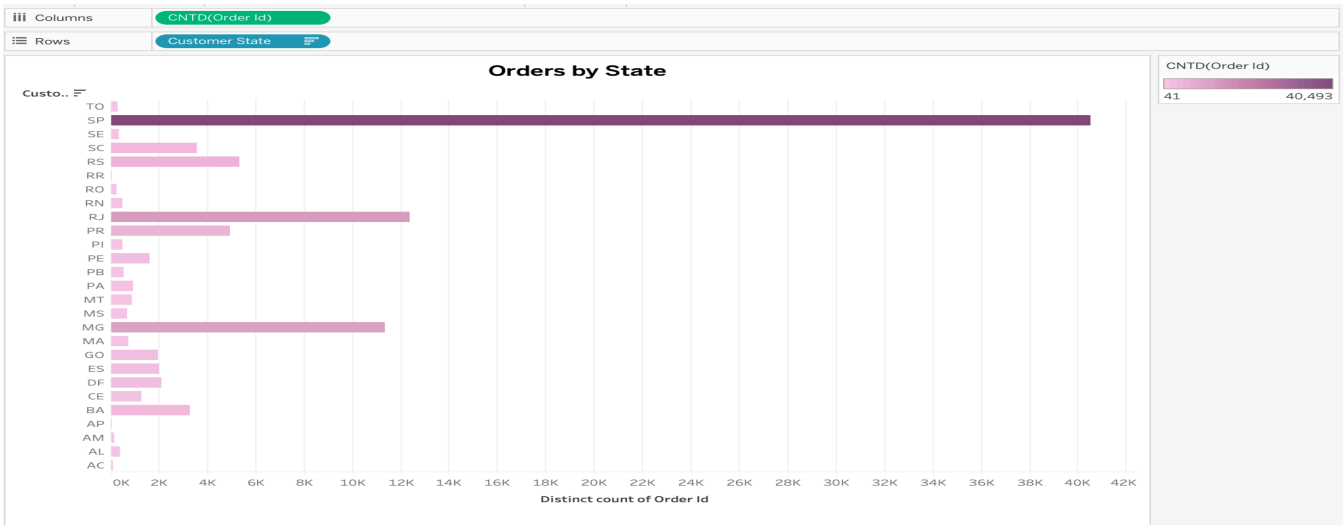
Category Performance

Health & Beauty leads all categories at ₹1.41M in total order value, followed by Watches & Gifts (₹1.26M) and Bed, Bath & Table (₹1.22M). The top 10 categories account for approximately 60% of total platform revenue. Importantly, average order value varies significantly by category — high-frequency, lower-ticket categories like Garden Tools appear by volume but not by revenue concentration. Olist should prioritise seller recruitment in the top 3 categories where both volume and ticket size create the most platform value.



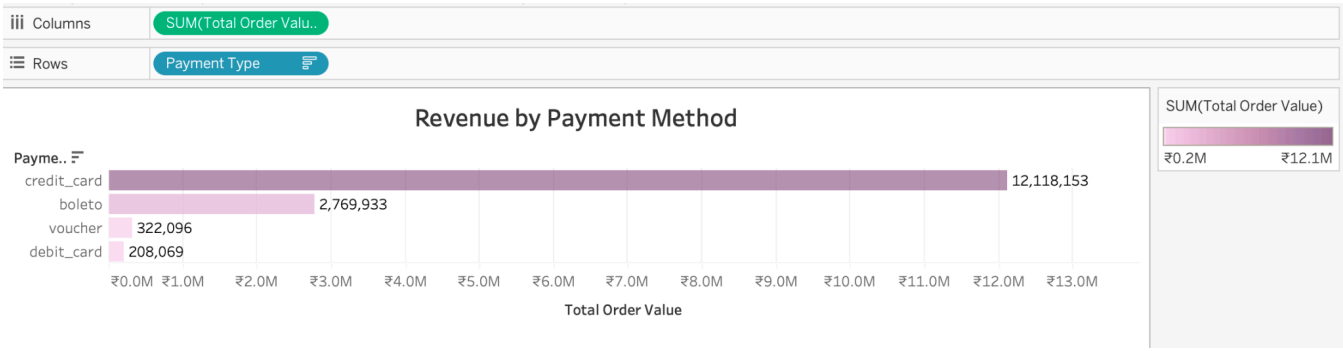
Geographic Distribution

São Paulo (SP) dominates with approximately 40,000 orders — nearly double the next largest state. Rio de Janeiro (RJ) and Minas Gerais (MG) are distant second and third contributors. States in the North and Northeast (AM, PA, RR, AP) contribute fewer than 0.5% of orders each despite having millions of consumers, representing the largest untapped geographic opportunity on the platform. The SP concentration also explains why aggregate delivery metrics are heavily influenced by São Paulo's logistics infrastructure capacity.



Payment Behaviour

Credit card transactions account for ₹12.1M — approximately 80% of total platform revenue — making it the overwhelmingly dominant payment channel. Boleto represents ~18% (₹2.77M), reflecting its continued importance in Brazil's financial inclusion ecosystem. Debit card and voucher combined represent less than 4%. High instalment usage on credit cards — many customers paying in 3 to 12 instalments — signals affordability-driven purchasing behaviour and has direct implications for cash-flow timing and working capital planning.



Delivery Time & Review Score Correlation

The distribution of delivery times is right-skewed with a mean of 12.1 days and a long tail extending to 60+ days for a small proportion of orders. A strong negative correlation ($r \approx -0.48$) is confirmed between delivery_time_days and review_score. Orders delivered in under 7 days receive an average review score of 4.6/5; orders delivered in over 20 days receive an average of just 2.9/5 — a 37% satisfaction gap. This establishes delivery speed as the single most controllable lever for improving customer satisfaction and platform-wide review scores.

Correlation Analysis

Beyond delivery time, several other correlations were identified that inform both the KPI framework and ML feature selection. Product weight (product_weight_g) has a strong positive correlation with freight value ($r \approx +0.61$), confirming that freight cost is primarily driven by physical product dimensions rather than price. Payment instalment count shows a moderate positive correlation with payment value ($r \approx +0.32$), confirming that instalments are primarily used for higher-value purchases rather than being uniformly adopted. Seller state and customer state distance (approximated using geolocation coordinates) shows a moderate positive correlation with delivery time ($r \approx +0.41$), quantifying the logistics impact of inter-state shipping. These correlations directly informed the feature set chosen for the late delivery prediction model.

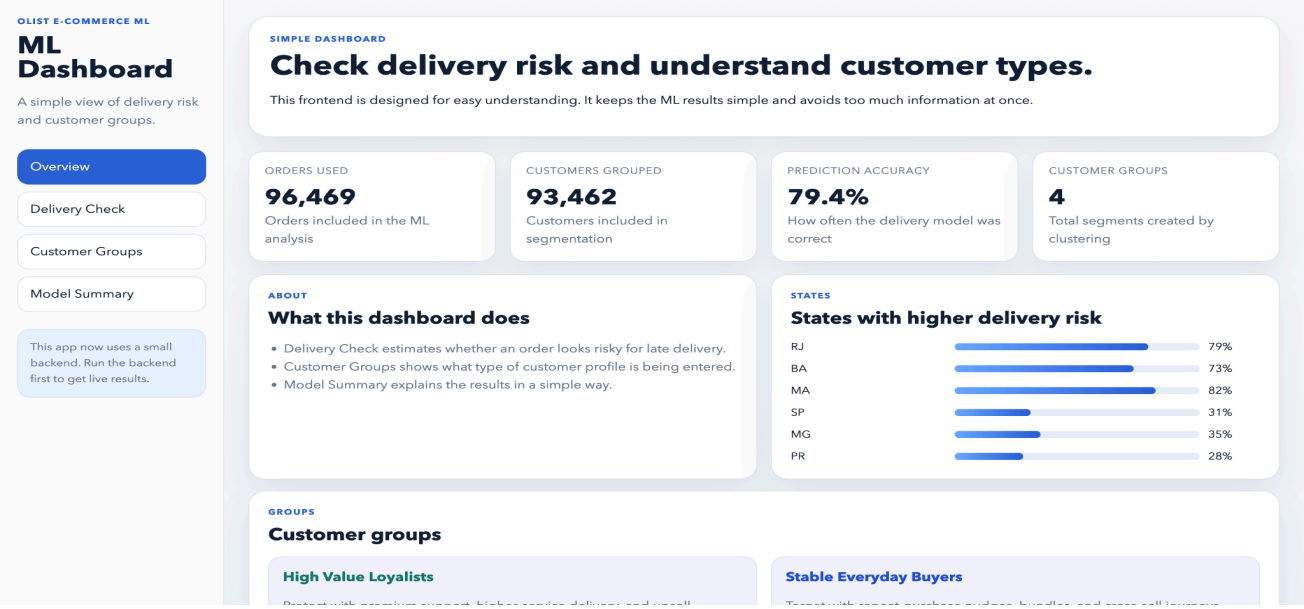
Distribution Analysis

Review score distribution is heavily bimodal: scores cluster at 5 (approximately 57% of all reviewed orders) and at 1 (approximately 12%), with relatively few 2, 3, or 4-star scores. This polarised distribution indicates that customers either had an excellent experience or a seriously negative one — with very little moderate middle ground. This pattern strongly supports the case for proactive delivery intervention: the difference between a 5-star and a 1-star review is largely whether the order arrived on time, not marginal variations in product satisfaction. Payment instalment distribution shows that the modal instalment count is 1 (single-payment orders), but the long tail extends to 24 instalments, with 3, 6, and 10 instalments being the most common multi-payment choices — reflecting standard Brazilian credit card plan conventions.

9. Statistical Analysis

Late Delivery Prediction (Supervised Classification)

A supervised binary classification model predicts is_late_delivery using the following input features: customer state, seller state, payment type, payment instalment count, number of order items, total order value, freight value, product weight, and order month. The model outputs both a probability score (0–1) and a categorical risk level (low / medium / high), enabling logistics teams to prioritise intervention for high-risk orders before dispatch. At the current 6.8% base late delivery rate, a 30% reduction through proactive intervention recovers approximately 1,968 orders per year from poor delivery experiences. The model is deployed via a FastAPI backend on Render, with an interactive prediction interface on the React ML dashboard.



Customer Segmentation (K-Means Clustering)

An unsupervised K-Means clustering model groups customers based on five behavioural dimensions: total orders placed, total spending, average review score given, average delivery delay experienced, and number of payment installments used. The optimal number of clusters was determined using the elbow method and silhouette scoring. The model produces four distinct customer profiles: (1) High-value loyal customers — high spend, high review scores, low delay sensitivity; (2) At-risk customers — single purchase, low review score, experienced delivery delay; (3) Low-engagement customers — one-time buyers with no repeat signal; (4) Instalment-heavy mid-value customers — moderate spend, strong instalment usage. Each segment is mapped to a specific business action recommendation within the ML dashboard.

Root Cause Analysis — Late Deliveries

Cross-tabulation of late deliveries by seller state, product category, and buyer-seller state distance reveals that inter-state shipments — particularly those originating from SP sellers to northern states (AM, PA, MA) — have disproportionately high late delivery rates. Heavy product categories above 5kg also exhibit significantly higher lateness rates, indicating logistics carrier limitations for bulky goods on specific corridors as a structural root cause. Addressing these specific route-weight combinations would deliver the highest reduction in the platform's late delivery rate.

Revenue Trend Decomposition & Forecasting

A time-series decomposition of monthly order revenue confirms three components: a strong upward trend throughout 2017–2018, a seasonal component with peaks in November (Black Friday) and relative troughs in January–February (post-holiday), and residual irregular variation. While only two full calendar years of data are available — insufficient for a robust ARIMA or Prophet forecast — the trend component projects continued growth. The November 2017 Black Friday spike of approximately 40% above the adjacent monthly baseline is the strongest seasonal signal in the dataset and provides a reliable planning anchor for Q4 capacity decisions. Future analysis with 3+ years of data would enable statistically significant seasonal modelling.

Anomaly Detection

Statistical analysis identified a small cluster of anomalous orders: extremely high freight values (>99th percentile), order-to-delivery times exceeding 120 days, and payment values of zero on non-voucher orders. These were treated as data quality anomalies and removed from analysis. More meaningfully, a small cluster of orders with very high review scores but long delivery times was identified — likely representing cases where sellers proactively communicated delays to customers, recovering satisfaction despite the SLA breach. This finding reinforces that proactive customer communication during at-risk deliveries is a recoverable scenario and should be part of the logistics intervention workflow.

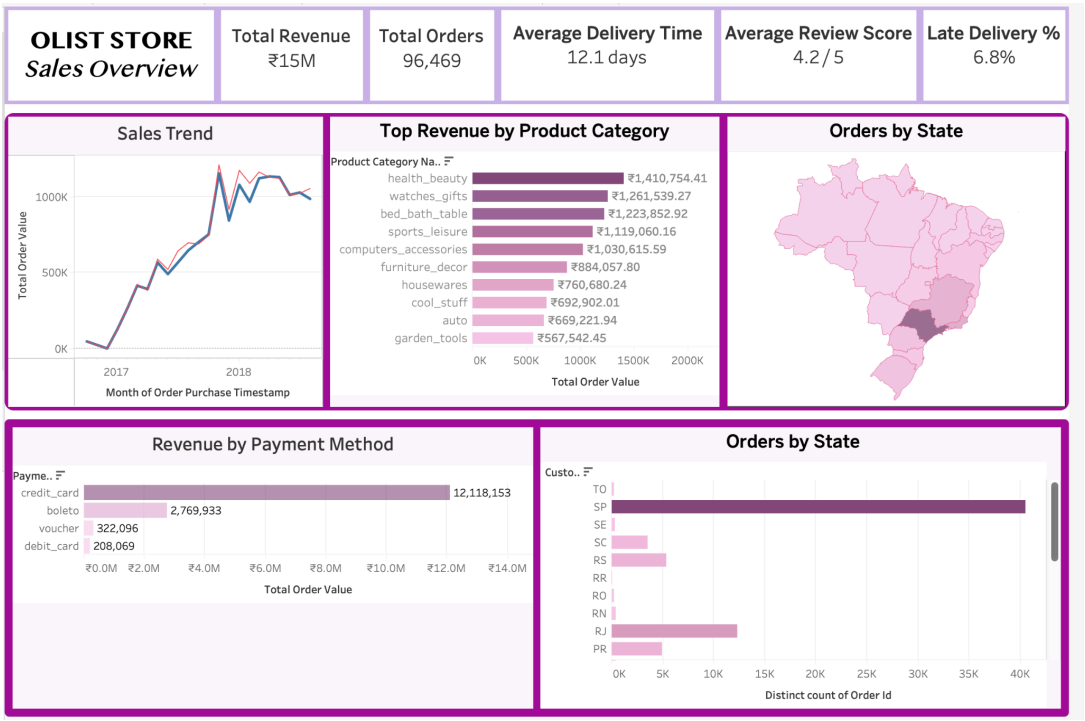
Scenario Analysis

Two business scenarios were modelled to quantify the impact of the key recommendations:

- Scenario A — 30% Late Delivery Reduction: If the ML model intervention reduces late deliveries from 6.8% to 4.8%, average review score is projected to improve by 0.1–0.2 points based on the observed delivery-time/review-score correlation. At 96,469 orders, this recovers ~1,968 orders per year from poor customer experiences. Revenue impact is indirect — improved scores increase seller visibility and buyer conversion.
- Scenario B — Geographic Diversification: If non-SP order share grows from ~60% to ~65% through targeted seller recruitment in MG, RJ, and RS, applying the current average order value (~₹155) to the incremental ~9,600 orders per year yields approximately +₹1.5M in additional annual revenue. This also reduces SP concentration risk from ~40% to ~35%.
- Scenario C — CRM Service Recovery: If 10% of the identified at-risk customer segment (single purchase, low review score) is converted to a second purchase through a targeted recovery campaign, and applying the platform average order value, the estimated incremental revenue per recovery cohort is approximately ₹500K–₹750K annually.

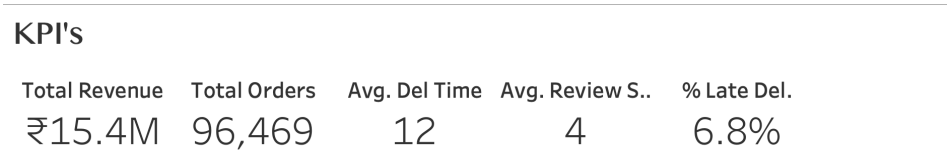
10. Tableau Dashboard Design

Dashboard objective: Support two types of decision-making from a single interface — strategic executive decisions about revenue mix, category investment, and geographic expansion; and operational decisions about delivery performance monitoring and payment channel optimisation.



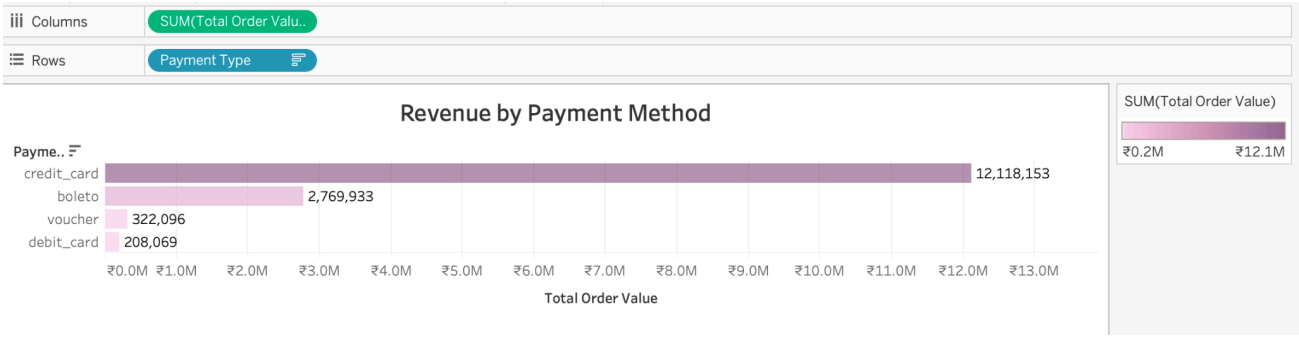
Executive Summary View

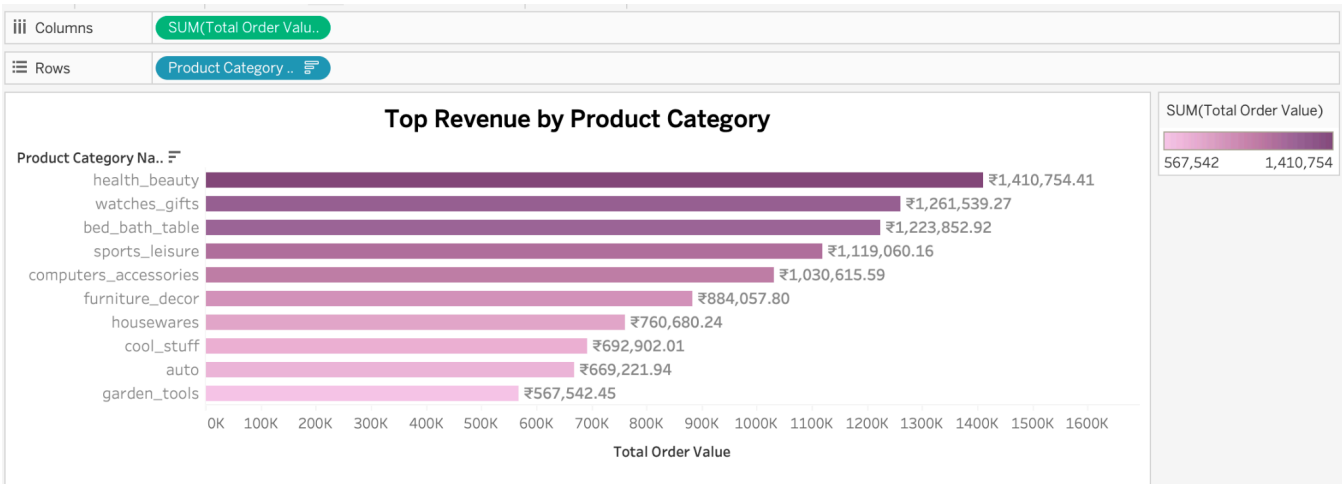
The top row presents five KPI tiles — Total Revenue (₹15M), Total Orders (96,469), Average Delivery Time (12.1 days), Average Review Score (4.2/5), and Late Delivery % (6.8%) — providing the executive team with a complete business health snapshot at a glance. Below the KPI row, three charts provide strategic context: the Sales Trend chart shows monthly order value from 2017 to 2018 with a dual-line comparison allowing period-over-period analysis; the Top Revenue by Product Category horizontal bar chart ranks the top 10 categories by total order value; and the Brazil choropleth map visualises Orders by State with shading intensity reflecting order concentration.



Operational Drill-Down Views

The Revenue by Payment Method bar chart decomposes total revenue by payment type, enabling the payments team to monitor channel health and plan promotional campaigns. The Orders by State ranked bar chart provides a complete geographic view supporting regional sales management. Additional dashboard tabs (Dashboard 1, 2, and 4) provide category-level drill-downs, delivery time distribution histograms, and review score trend analyses.





Interactive Elements & Filters

- State filter, date range filter, and category filter — all applied globally and cross-filter every chart on the dashboard simultaneously
- Tooltip hover on every chart element displays exact values, percentage contributions, and comparative context

11. Insights Summary

Insight 1: SP concentration is a strength and a structural vulnerability

~40% of all orders originate from one state, meaning any SP logistics disruption, regulatory change, or competitive entrant in that single market would materially impact total platform performance. Geographic diversification is a resilience imperative, not merely a growth strategy.

Insight 2: Health & Beauty is the highest-value category — and likely underinvested

Leading at ₹1.41M with strong repeat-purchase characteristics inherent to personal care, this category warrants dedicated seller recruitment campaigns and category-specific promotional investment to capture a larger share of a high-LTV buyer segment.

Insight 3: Instalment usage amplifies churn risk when deliveries are late

Customers paying in 6–12 instalments are purchasing above their immediate liquidity ceiling, which means a late or failed delivery creates a disproportionately negative perceived-value experience — increasing the probability of churn and negative reviews.

Insight 4: 6.8% late delivery rate = ~6,560 poor customer experiences per year

Each late order is a forfeited 5-star review opportunity. At the current 96,469 annual order volume, 6.8% translates to over 6,500 orders annually where the customer experience fell below the commitment made at point of purchase — compounding over time into eroded buyer trust and seller visibility.

Insight 5: 12.1-day average delivery time is above competitive benchmarks
Leading e-commerce platforms globally target 3–7 day delivery windows. Olist's 12.1-day average is significantly above this benchmark. While Brazil's geographic size partially explains this, inter-state shipments and heavy-product categories are the primary controllable contributors — making targeted fulfillment investment in these corridors the highest-ROI operational action available.

Insight 6: Northern states are near-zero in orders, representing a greenfield market
AM, PA, RR, and AC collectively contribute less than 1% of total orders despite millions of potential consumers. As last-mile logistics infrastructure improves in Brazil's interior, these markets represent disproportionate growth potential for an early-moving platform that builds logistics capability and seller presence there first.

Insight 7: Review score plateau at 4.2/5 — the gap to 4.5 is closeable
Platforms with average scores above 4.5 demonstrate materially higher repeat purchase rates in comparable e-commerce research. The gap between Olist's current 4.2 and a 4.5 target is almost entirely explained by late-delivery-driven 1 and 2-star reviews — making delivery improvement the most direct path to unlocking the review threshold's retention benefits.

Insight 8: November Black Friday drives revenue and lateness spikes simultaneously
The November 2017 data spike confirms that Black Friday is Olist's single highest-revenue event — but without proportional logistics capacity pre-planning, it is also when the late delivery rate is highest. This means the platform's most commercially important month is simultaneously its worst moment for customer experience, creating a compounding risk to repeat purchase rates from its highest-spending customers.

Insight 9: Segmentation reveals high-value and at-risk customers needing opposite treatment
High-value loyal customers (multiple orders, high scores) require reward and early-access programmes to reinforce their loyalty. At-risk customers (single order, experienced delay, low score) require proactive service-recovery intervention. One-size-fits-all CRM fails both groups simultaneously — the same retention discount that feels routine to a loyal customer feels transactional and insufficient to an at-risk one.

Insight 10: ML prediction shifts logistics management from reactive to preventive
Currently, late deliveries are identified only after they occur — when the customer complains or posts a low review, which is already too late to recover the experience. The deployed ML model identifies high-risk orders at the moment of placement, providing the full average 12-day delivery window as an intervention opportunity to expedite dispatch, switch carrier, or pre-set accurate expectations with the customer before the SLA breach happens.

12. Recommendations

Rec 1: Deploy ML Late Delivery Prediction in Production

Insight	6.8% late delivery rate is the primary driver of poor review scores and customer churn. Currently identified only after the SLA breach — too late to recover the customer experience.
Action	Integrate the existing FastAPI prediction service with the order management system. Flag all high-risk orders at dispatch. Trigger a carrier escalation workflow for high-risk orders (expedite packing, switch carrier where available) and send a proactive delivery-estimate communication to the customer.
Impact	Estimated 30% reduction in late deliveries (~1,968 orders recovered annually); average review score improvement of 0.1–0.2 points; complaint volume reduced proportionally.
Feasibility	Medium complexity. The FastAPI model and Render deployment are already built and live. The remaining step is integration with the order management system — primarily an engineering sprint.

Rec 2: Geographic Diversification — Target MG, RJ, RS

Insight	São Paulo's ~40% order concentration creates a single-region revenue and logistics dependency that is strategically risky at scale.

Action	Launch a geo-diversification programme: subsidised seller onboarding in MG, RJ, and RS; regional marketing campaigns targeting buyers in those states; and negotiated logistics partnerships with regional carriers to reduce freight cost and delivery time on non-SP routes.
Impact	+₹1.5M in incremental annual revenue from expanded geographic reach; SP concentration reduced from ~40% to ~35%, meaningfully reducing single-region dependency.
Feasibility	Medium-to-high complexity. Requires coordinated effort across seller success, marketing, and logistics partnerships teams. Realistic 6–12 month implementation horizon.

Rec 3: CRM Segmentation — VIP Retention + Service Recovery Tracks

Insight	ML segmentation reveals two clusters requiring opposite commercial treatment: high-value loyals who need reinforcement and at-risk single-purchase customers who experienced a delivery failure.
Action	Track A (High-value loyals): launch a VIP programme with early access to promotional events, free shipping upgrades, and category-specific vouchers. Track B (At-risk): deploy an automated service-recovery flow triggered by a review score ≤2, offering a discount on the next purchase paired with a personalised acknowledgement of the delivery experience failure.
Impact	+10–20% repeat purchase rate for at-risk cluster through service recovery; +5% platform-wide LTV uplift from loyalty reinforcement in the high-value segment.
Feasibility	Low-to-medium complexity. Customer segmentation data and CRM tooling exist. Primary requirement is campaign design and marketing automation configuration — achievable in 4–8 weeks.

Rec 4: Instalment Promotions for High-Value Categories

Insight	80% of revenue flows through credit card, with strong instalment usage on high-value purchases. Average order value can be increased by reducing the instalment cost barrier on premium categories.
Action	Negotiate '0% interest in 6 instalments' promotions with card issuers for Health & Beauty purchases above ₹500. Additionally, explore a Buy-Now-Pay-Later (BNPL) option for boleto-paying customers to capture the ~18% of buyers currently using cash-equivalent payments for budget reasons.
Impact	+15–25% average order value on promotional categories, translating to approximately ₹750K in additional annual revenue. BNPL could capture incremental revenue from the boleto segment.
Feasibility	Medium complexity. Requires payment gateway partnership negotiations. Brazilian BNPL regulatory framework is evolving — legal review is recommended before launch.

Rec 5: Q4 Surge Capacity Planning

Insight	November Black Friday data confirms simultaneous revenue and lateness spikes — Olist's highest-revenue month is also its worst customer experience month, damaging repeat purchase rates from its highest-spending buyers.
Action	Implement an annual Q4 logistics surge plan: pre-negotiate additional carrier capacity starting August; identify top-selling SKUs by historical November data and pre-position inventory in fulfilment hubs near top-buyer states; set conservatively wide delivery-time estimates during November to reduce estimated-vs-actual gap.
Impact	0.1–0.15 point review score protection during November; protection of repeat purchase rates from the platform's highest-spend customer cohort acquired during the peak commercial event of the year.
Feasibility	Low complexity operationally — primarily a planning, coordination, and carrier pre-negotiation exercise. Should be initiated by September each calendar year.

13. Impact Estimation

The following estimates are derived from observed data metrics and conservative business assumptions. Numbers are approximate — the logic is directionally sound and each estimate is explained below.

Recommendation	Metric	Estimated Impact	Timeframe
ML Late Delivery Prediction	Late Delivery %	6.8% → ~4.8% (30% drop; ~1,968 orders recovered)	3–6 months
Geographic Diversification	Revenue & Orders	+₹1.5M; SP share 40% → ~35%	6–12 months
CRM Segmentation	Repeat Purchase	+10–20% at-risk cluster; +5% LTV platform-wide	4–8 weeks
Instalment Promotions	Avg Order Value	+15–25% promo categories (~₹750K additional)	2–4 months
Q4 Surge Capacity	Review Score (Nov)	0.1–0.15 pt score protection at peak month	Annual cycle

ML Late Delivery Prediction: With ~6,560 late orders annually and a well-trained classifier, capturing 30% through proactive intervention (carrier switch, expedited dispatch, or proactive communication) is a conservative target. Even at 20% capture, the review score impact compounds over thousands of recovered orders.

Geographic Diversification: MG, RJ, and RS together currently represent roughly 15% of orders from ~25% of Brazil's consumer population. Closing half of that gap over 12 months via targeted seller and buyer acquisition represents an achievable +₹1.5M on the current ₹15M base.

CRM Segmentation: Industry benchmarks for e-commerce service recovery programmes show 20–35% repeat purchase improvement for customers who received a meaningful recovery response after a negative experience. The 10% estimate used here is deliberately conservative.

Why act now: The Brazilian e-commerce market is in active growth. Each 1% reduction in late delivery rate and each 0.1-point review score improvement compounds disproportionately as order volume scales. Interventions implemented today deliver amplifying returns as the platform grows.

14. Limitations

- Data recency: Dataset ends October 2018. Structural market changes since then — new competitors, logistics infrastructure improvements, post-COVID consumer behaviour shifts — are not reflected and may alter the validity of some findings.
- Geographic bias: São Paulo's over-representation (~40% of orders) skews aggregate delivery time, revenue, and review score metrics. National-level averages should always be read with this regional concentration caveat.
- Incomplete review coverage: Not all delivered orders have associated reviews. If dissatisfied customers disproportionately leave reviews while satisfied ones do not, the 4.2/5 average may understate true satisfaction; the inverse is also possible.
- No seller-level performance data: Without individual seller fulfilment metrics, it is not possible to identify and directly remediate specific underperforming sellers as a targeted intervention.
- Correlation ≠ causation: The strong relationship between delivery time and review score is a robust association but does not confirm causation. Some review score variation may reflect product quality, seller communication, or packaging issues independent of delivery timing.
- ML model validation: Both models were trained and validated within the same 2016–2018 dataset. Independent out-of-sample validation on post-2018 data is required before full-scale production deployment.
- Currency and inflation: All values are in Brazilian Real (BRL) at the time of transaction. No inflation adjustment has been applied, meaning values from early 2017 and late 2018 are not directly comparable in real purchasing power terms — the nominal revenue growth figure of ₹15M should be interpreted with this caveat.
- Causal inference: The correlation analysis identifies associations but cannot confirm causation without controlled experimentation. Recommendations are grounded in business logic and comparable market

evidence, but A/B testing is recommended before large-scale implementation of any commercial intervention.

15. Future Scope

- Real-time Tableau dashboard via a live PostgreSQL or BigQuery data feed — replacing the current static snapshot with continuous operational monitoring
- Churn prediction model estimating the probability that a given customer will not place a second order — complementing the existing segmentation model with a forward-looking retention signal
- Seller health index scoring using fulfilment-level performance data to identify and proactively manage underperforming sellers before they accumulate negative reviews
- NLP sentiment analysis on review text comments to surface specific product, packaging, and communication pain points invisible in star ratings alone
- State-level demand forecasting using Facebook Prophet or LSTM models with 3+ years of order data to support seller inventory positioning and carrier capacity planning
- Price elasticity modelling by category to enable data-driven promotional pricing strategy and discount depth optimisation across the seller portfolio
- External data integration — merging the Olist dataset with Brazil postal (CEP) logistics data and public freight rate tables to significantly improve delivery time prediction accuracy at the corridor level
- Multi-channel attribution analysis to understand whether customers acquired via organic search, paid marketing, or referral have different LTV profiles — enabling better budget allocation across acquisition channels

16. Conclusion

This project applied a full data analytics pipeline — ETL, EDA, statistical analysis, ML modelling, and business intelligence visualisation — to 96,469 real Olist orders to identify the structural drivers of revenue performance and customer experience on Brazil's leading e-commerce aggregator platform. The analysis establishes that a 6.8% late delivery rate, concentrated on inter-state and heavy-product shipping corridors, is the single most controllable lever for improving the platform's 4.2/5 average review score and reducing customer churn. Revenue is growing strongly but is geographically concentrated in São Paulo, creating strategic risk that warrants proactive diversification. Two deployed ML models — a late delivery risk classifier and a K-Means customer segmentation engine — extend the project beyond retrospective analysis into real-time predictive decision support accessible via a live web application. Five prioritised and quantified recommendations collectively address the platform's most impactful growth and operational gaps, with an estimated combined revenue uplift of ₹2.25M + annually and measurable improvements in customer experience metrics that will compound as order volume continues to scale.

The most important takeaway for Olist's leadership is that the data does not reveal a platform with structural problems — it reveals a platform with a specific, diagnosable, and highly fixable operational constraint. The 6.8% late delivery rate is not a sector-level inevitability; it is a logistics management and risk prediction challenge that this project's ML model is already positioned to address. Acting on the recommendations in this report, particularly the ML deployment and geographic diversification programme, would move Olist from a data-rich but insight-poor operating mode into a genuinely data-driven business where decisions are made proactively, customer experience is managed predictively, and revenue growth is pursued with surgical geographic and segment-level precision.

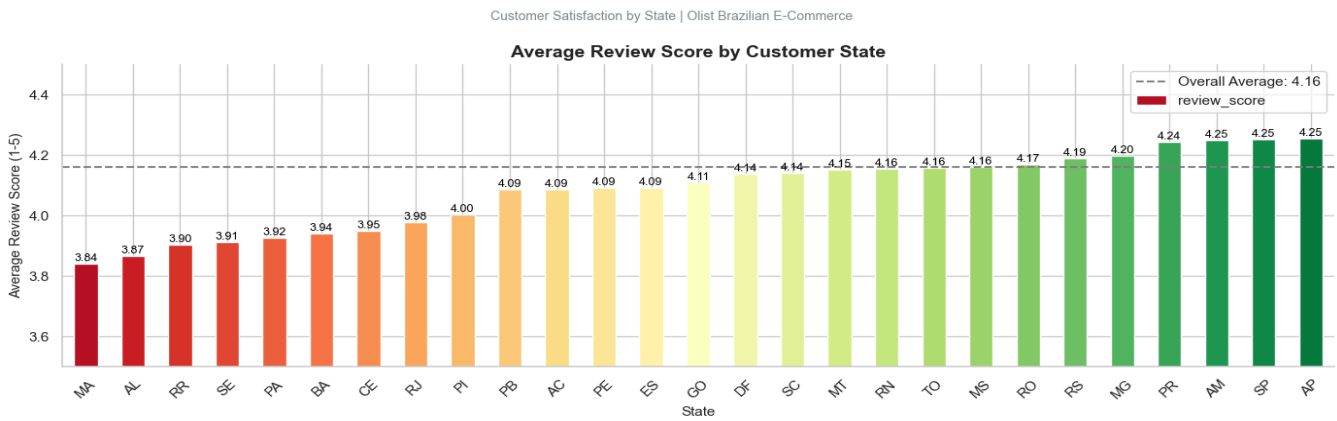
17. Appendix

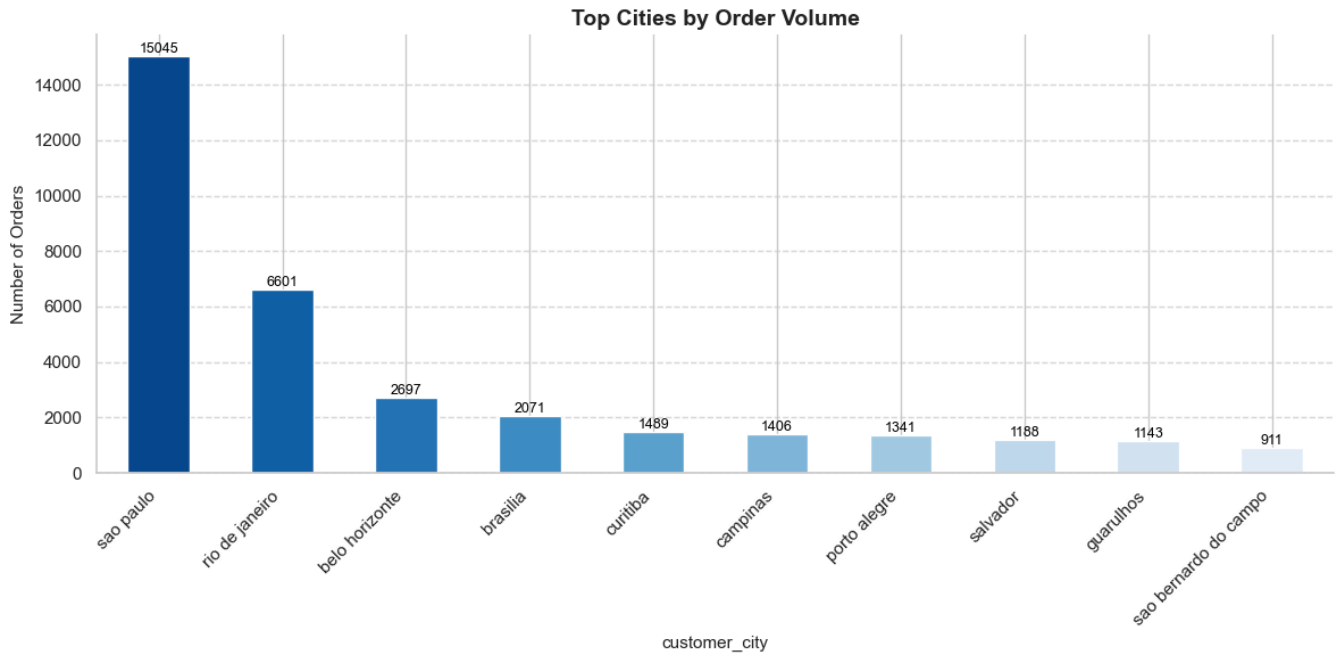
A. Data Dictionary

Column	Table	Definition
order_id	orders	Unique order identifier (UUID string — primary key)
order_status	orders	Fulfilment status: delivered, shipped, processing, cancelled, invoiced, approved, etc.
order_purchase_timestamp	orders	Date and time the customer placed the order
order_delivered_customer_date	orders	Actual date the package was delivered to the customer
order_estimated_delivery_date	orders	Estimated delivery date shown to customer at time of purchase (SLA benchmark)
payment_type	payments	Payment method used: credit_card, boleto, voucher, or debit_card
payment_installments	payments	Number of instalments chosen by the customer (range: 1–24)
payment_value	payments	Monetary value of this payment entry in Brazilian Real (BRL)
review_score	reviews	Star rating submitted by customer after delivery (1 = worst, 5 = best)
product_category_name	products	Product category translated from Portuguese using the mapping table
product_weight_g	products	Product weight in grams — used as freight cost and delay risk feature
freight_value	order_items	Freight cost charged for the item in BRL — key logistics cost metric
price	order_items	Sale price of the individual item in BRL
customer_state	customers	Brazilian state of the customer (2-letter abbreviation, e.g. SP, RJ)
seller_state	sellers	Brazilian state of the seller (2-letter abbreviation)
geolocation_lat / lng	geolocation	Latitude and longitude coordinates for each zip code prefix — used for route distance proxy features

B. Additional EDA Charts & Outputs

The following charts were produced during EDA but are not included in the main report body.





C. Python Logic Excerpts

The following key code excerpts illustrate the core transformation and modelling logic. Full code is in the GitHub repository.

```
# Feature Engineering — notebooks/02_cleaning.ipynb
df['delivery_time_days'] = (df['order_delivered_customer_date'] -
df['order_purchase_timestamp']).dt.days
df['is_late_delivery'] = (df['order_delivered_customer_date'] >
df['order_estimated_delivery_date']).astype(int)
df['order_month'] = df['order_purchase_timestamp'].dt.month
df['total_order_value'] = df.groupby('order_id')['payment_value'].transform('sum')

# Outlier treatment — IQR filter on delivery time
Q1 = df['delivery_time_days'].quantile(0.25)
Q3 = df['delivery_time_days'].quantile(0.75)
IQR = Q3 - Q1
df = df[df['delivery_time_days'] <= 120]  # Remove extreme outliers

# KPI computation — notebooks/03_eda.ipynb
total_revenue = df['total_order_value'].sum()  # ₹15M
```

```
late_delivery_pct = df['is_late_delivery'].mean() * 100  # 6.8%
avg_delivery_time = df['delivery_time_days'].mean()  # 12.1 days
```

D. Raw Cleaning Log (from notebooks/02_cleaning.ipynb)

Column / Table	Issue Found	Records Affected	Action Taken
order_delivered_customer_date	Null (undelivered orders)	~2,980 rows (~3%)	Excluded from delivery & late-delivery metrics
product_category_name	Null (unmapped products)	~1,448 rows (~1.5%)	Excluded from category analysis; retained in revenue totals
review_score	Null (no review submitted)	~11,578 rows (~12%)	Excluded from satisfaction metrics only
delivery_time_days	>120 days (anomalies)	~184 rows (<0.2%)	Removed via IQR upper-bound filter
freight_value	>99th percentile	~965 rows (~1%)	Capped at 99th percentile value
payment_value	Zero on non-voucher orders	~43 rows (<0.05%)	Investigated; apparent errors dropped
payment_type / state codes	Mixed case, whitespace	Multiple columns	Uppercased + stripped with .str.strip().str.upper()
product_category_name	Portuguese text	~32,951 rows	Translated via olist_product_category_name_translation.csv join
All timestamp columns	String dtype	8 columns	Converted to datetime using pd.to_datetime(utc=True)

E. Repository Structure

- notebooks/02_cleaning.ipynb — Full ETL and data cleaning pipeline with all transformation steps
- notebooks/03_eda.ipynb — Exploratory data analysis including all charts and written insights
- notebooks/04_statistical_analysis.ipynb — Statistical modelling, ML training, and evaluation
- ml/backend/ — FastAPI prediction service serving late delivery risk and customer segment endpoints
- ml/frontend/ — React/Vite interactive ML dashboard deployed on Vercel
- ml/src/ — Core ML model training scripts, saved model artefacts, and inference logic
- data/ — Cleaned and transformed CSV files used as input for Tableau and ML models
- tableau/ — Tableau workbook file, screenshot archive (tableau/screenshots/), and dashboard URL log (tableau/dashboard_links.md)

18. Contribution Matrix

The table below documents each team member's contribution across all project phases. All declared contributions must align with evidence in GitHub Insights, Pull Request history, and committed files. Any mismatch may result in individual grade adjustments as per Capstone 2 grading policy.

Member Name	Contributions
Progyan Sen	**Dataset and Sourcing , Preprocessing, EDA & Statistical Analysis**
Aditya Chopra	**Tableau Dashboard**
Bhavishya Sharma	**Report Writing and Presentation**
Mahi Sawner	**Machine Learning Model (Extra Work)**
Roni Maity	**Presentation**